# Sensor Data: test files

15 CSV files in the 'Sensor Data' folder, each built to test one situation. Expected results below were produced by running each file through the real pipeline with a dry forecast (0 mm rain), 1 acre, drip, clay loam soil. With rain in the live forecast, 'irrigate' can become 'wait for rain': that is the rain rule working.

How to use: Farms page → add a farm with the same crop → Upload my sensor CSV. Or Dose Calculator → Sensor data: Upload a CSV file (choose the crop in the form; the form's crop is used). Or the API: curl -X POST http://127.0.0.1:8000/api/advice -F csv=@"Sensor Data/01_tomato_dry_soil.csv"

Columns: timestamp, moisture (%), N, P, K, pH, temperature (°C), humidity (%), rainfall (mm), crop.

### 01_tomato_dry_soil.csv

**RED · irrigate then fertilize**

Tomato, soil drying from 31% to 22%, low N/P/K (the challenge sample).
Rows: 8 · Crop: tomato · Confidence: 100% HIGH
Water: 35,972 L today · Fertilizer: urea 19.8 kg, DAP 80.6 kg, MOP 23.1 kg

### 02_tomato_healthy.csv

**GREEN · no action**

Tomato, moisture steady above the trigger, nutrients above target.
Rows: 8 · Crop: tomato · Confidence: 100% HIGH
Water: 0 L today · Fertilizer: none needed

### 03_tomato_waterlogged.csv

**RED · drainage warning**

Tomato, soil above field capacity (too wet).
Rows: 8 · Crop: tomato · Confidence: 100% HIGH
Water: 0 L today · Fertilizer: none needed

### 04_sensor_flatline.csv

**RED · inspect sensor**

Moisture sensor stuck at the same value for the last 5 readings.
Rows: 8 · Crop: tomato · Confidence: 40% MEDIUM
Doses: none (sensor fault)
Confidence lowered by: flatline sensor reading; sensor fault: confidence capped

### 05_sensor_out_of_range.csv

**RED · inspect sensor**

Moisture sensor reports an impossible 140%.
Rows: 8 · Crop: tomato · Confidence: 40% MEDIUM
Doses: none (sensor fault)
Confidence lowered by: sensor spike detected; sensor fault: confidence capped

### 06_sensor_spike.csv

**RED · irrigate then fertilize**

One spike reading in otherwise smooth data.
Rows: 8 · Crop: tomato · Confidence: 95% HIGH
Water: 35,972 L today · Fertilizer: urea 19.8 kg, DAP 80.6 kg, MOP 23.1 kg
Confidence lowered by: sensor spike detected

### 07_data_gap.csv

RED · irrigate then fertilize

Readings missing for 4 days in the middle.
Rows: 8 · Crop: tomato · Confidence: 90% HIGH
Water: 35,972 L today · Fertilizer: urea 19.8 kg, DAP 80.6 kg, MOP 23.1 kg
Confidence lowered by: data gap > 2x median interval

### 08_missing_npk.csv

RED · irrigate

Only moisture, pH and climate: no N/P/K sensor.
Rows: 8 · Crop: tomato · Confidence: 80% HIGH
Water: 35,972 L today · Fertilizer: not computed (no N/P/K)
Confidence lowered by: missing NPK readings

### 09_only_3_days.csv

RED · irrigate then fertilize

Only 3 days of readings.
Rows: 3 · Crop: tomato · Confidence: 85% HIGH
Water: 35,972 L today · Fertilizer: urea 19.8 kg, DAP 80.6 kg, MOP 23.1 kg
Confidence lowered by: < 5 days of data

### 10_paddy_ponded.csv

GREEN · no action

Rice field kept flooded: moisture at saturation.
Rows: 8 · Crop: rice · Confidence: 100% HIGH
Water: 0 L today · Fertilizer: none needed

### 11_paddy_drying.csv

RED · irrigate then fertilize

Rice field: ponded water has gone, soil drying below field capacity.
Rows: 8 · Crop: rice · Confidence: 100% HIGH
Water: 337,238 L today · Fertilizer: urea 20.3 kg, DAP 60.4 kg, MOP 11.6 kg

### 12_maize_needs_fertilizer.csv

YELLOW · fertilize

Maize: moisture fine for maize, nutrients low.
Rows: 8 · Crop: maize · Confidence: 100% HIGH
Water: 0 L today · Fertilizer: urea 55.3 kg, DAP 120.9 kg, MOP 23.1 kg

### 13_cotton_dry.csv

RED · irrigate then fertilize

Cotton: very dry soil, below cotton's lower trigger.
Rows: 8 · Crop: cotton · Confidence: 100% HIGH
Water: 35,972 L today · Fertilizer: urea 20.3 kg, DAP 60.4 kg, MOP 11.6 kg

### 14_banana_dataset_crop.csv

RED · irrigate then fertilize

Banana (target from the Kaggle dataset).
Rows: 8 · Crop: banana · Confidence: 90% HIGH
Water: 35,972 L today · Fertilizer: urea 0.0 kg, DAP 165.2 kg, MOP 11.6 kg
Confidence lowered by: fertilizer target from crop dataset, not a TNAU recommendation

### 15_hourly_readings.csv

RED · irrigate

Hourly readings for 7 days (168 rows), slowly drying.
Rows: 168 · Crop: tomato · Confidence: 100% HIGH
Water: 35,972 L today · Fertilizer: none needed